i want create the GEnI ML and Deep lienaing repo on firhub

I am android developoer more specifiucally mobile devloper whoshas 12 years o exprenince

BNow i am leanirng gen ai and ml and deep lenaing

what is want is

1 - Repository name
2 - How the repsosity should be strucure
3. It will contains questions and concept , POC and all

Create one **learning + portfolio monorepo** rather than many scattered repos. A strong name is `genai-ml-dl-journey`—it clearly says what the repo covers, works well in search, and sounds professional for a client-facing GitHub profile. GitHub recommends a README, license, and clear repository documentation so visitors can understand and navigate the project.[1–1][1_2]

1. Repository name

Recommended: `genai-ml-dl-journey`

Other good options:

Name	Best for
`genai-ml-dl-journey`	General learning, concepts, POCs, interview prep
`ai-mobile-engineer`	Positioning yourself as a mobile developer moving into AI
`genai-mobile-lab`	Emphasizing GenAI + Android/iOS use cases
`ml-dl-genai-notes`	More notes/interview-question focused
`applied-genai-pocs`	More project/POC focused

Use lowercase, hyphens, and no spaces. Avoid names like `test`, `learning`, or `AI-repo` because they do not communicate your specialization.

2. Recommended structure

Since you are learning ML, Deep Learning, and GenAI while also being a 12-year mobile developer, structure the repo by **skill area**, with a separate section for mobile-AI POCs. This keeps concepts, interview questions, and runnable experiments easy to find.

```
genai-ml-dl-journey/
│
├── README.md
├── LICENSE
├── .gitignore
├── requirements.txt
├── pyproject.toml
├── .env.example
│
├── 00-roadmap/
│   ├── learning-roadmap.md
│   ├── weekly-plan.md
│   └── progress-tracker.md
│
├── 01-python-for-ai/
│   ├── concepts/
│   ├── questions/
│   └── notebooks/
│
├── 02-math-for-ml/
│   ├── linear-algebra/
│   ├── statistics/
│   ├── probability/
│   └── questions/
│
├── 03-machine-learning/
│   ├── 01-data-preprocessing/
│   ├── 02-supervised-learning/
│   ├── 03-unsupervised-learning/
│   ├── 04-model-evaluation/
│   ├── 05-feature-engineering/
│   ├── concepts/
│   ├── questions/
│   └── notebooks/
│
├── 04-deep-learning/
│   ├── 01-neural-networks/
│   ├── 02-cnn/
│   ├── 03-rnn/
│   ├── 04-transformers/
│   ├── 05-pytorch/
│   ├── 06-tensorflow/
│   ├── concepts/
│   ├── questions/
│   └── notebooks/
│
├── 05-generative-ai/
│   ├── 01-llm-fundamentals/
│   ├── 02-prompt-engineering/
│   ├── 03-embeddings/
│   ├── 04-vector-databases/
│   ├── 05-rag/
│   ├── 06-agents/
│   ├── 07-fine-tuning/
│   ├── 08-evaluation-and-safety/
│   ├── concepts/
```

```
│   ├── questions/
│   └── notebooks/
│
├── 06-mobile-ai/
│   ├── android-on-device-ml/
│   ├── ios-core-ml/
│   ├── tensorflow-lite/
│   ├── mediapipe/
│   ├── gemini-api/
│   ├── openai-api/
│   ├── local-llms/
│   └── pocs/
│
├── 07-pocs/
│   ├── 01-rag-document-assistant/
│   ├── 02-ai-support-agent/
│   ├── 03-mobile-app-review-analyzer/
│   ├── 04-voice-assistant/
│   ├── 05-image-captioning-mobile-app/
│   └── 06-tender-document-workflow/
│
├── 08-interview-prep/
│   ├── ml-interview-questions.md
│   ├── dl-interview-questions.md
│   ├── genai-interview-questions.md
│   ├── system-design-ai.md
│   └── mobile-ai-interview-questions.md
│
├── 09-projects/
│   ├── project-ideas.md
│   ├── case-studies/
│   └── architecture-diagrams/
│
├── docs/
│   ├── setup-guide.md
│   ├── coding-standards.md
│   └── resources.md
│
└── .github/
    └── workflows/
        └── ci.yml
```

This follows good ML repository practice: separate notebooks/experiments from reusable source code, keep configuration separate, avoid committing large datasets or secrets, and document setup and usage in the README.[1–3][1_4][^1_5]

3. What each section should contain

`00-roadmap`

Track your journey from mobile developer to AI engineer.

Include:

- Current skills: Android, Kotlin, Java, mobile architecture, testing.
- Target skills: Python, ML, Deep Learning, LLMs, RAG, agents, on-device AI.
- Monthly milestones.
- Completed topics.
- Portfolio projects planned.

`01-python-for-ai`

Your first technical foundation.

Add:

- Python basics for AI.
- NumPy, Pandas, Matplotlib.
- Virtual environments.
- Jupyter Notebook usage.
- API calls.
- JSON handling.
- Async basics.

Example files:

```
01-python-for-ai/
├── concepts/
│   ├── 01-python-basics.md
│   ├── 02-numpy.md
│   └── 03-pandas.md
├── questions/
│   └── python-interview-questions.md
└── notebooks/
    └── 01-numpy-pandas-basics.ipynb
```

`03-machine-learning`

Use this for classical ML.

Cover:

- Regression, classification, clustering.
- Train/test split.

- Overfitting and underfitting.
- Precision, recall, F1 score, ROC-AUC.
- Scikit-learn.
- Feature engineering.
- Model deployment basics.

Since you already explored precision, recall, and F1 score, keep a dedicated `04-model-evaluation/` folder with small notebooks and explanations.

`04-deep-learning`

Use this for neural networks and model training.

Cover:

- Perceptrons and neural networks.
- Activation functions.
- Backpropagation.
- CNNs for images.
- RNNs and sequence models.
- Transformers.
- PyTorch and TensorFlow/Keras.
- Transfer learning.

`05-generative-ai`

This should become the strongest part of your repository.

Cover:

- LLM fundamentals: tokens, context window, temperature, embeddings.
- Prompt engineering.
- RAG: document loading, chunking, embedding, vector search, generation.
- Vector databases: FAISS, Chroma, Pinecone, Qdrant.
- Agents and tool calling.
- Fine-tuning versus RAG.
- Evaluation, hallucinations, safety, cost, and latency.

For client-facing credibility, every GenAI POC should include a problem statement, architecture, sample input/output, limitations, and cost/control considerations—not only code.

`06-mobile-ai`

This is your **differentiator**. Most ML learners do not have 12 years of mobile experience.

Add mobile-specific AI topics:

- On-device ML with TensorFlow Lite / LiteRT.
- MediaPipe for vision and audio.
- Core ML for iOS.
- Gemini API integration in Android.
- OpenAI-compatible APIs in mobile apps.
- Local LLMs on mobile.
- Mobile RAG.
- Privacy-first AI features.
- Offline AI features.
- AI-powered mobile UX.

Example POCs:

- Android app that summarizes user reviews.
- Offline image classifier using TensorFlow Lite.
- Voice-to-task mobile assistant.
- Document scanner plus RAG assistant.
- AI chatbot with on-device fallback.
- Mobile app that classifies support tickets.

`07-pocs`

Keep each POC self-contained:

```
07-pocs/
└── 01-rag-document-assistant/
    ├── README.md
    ├── architecture.png
    ├── requirements.txt
    ├── .env.example
    ├── src/
    │   ├── loader.py
    │   ├── chunking.py
    │   ├── embeddings.py
    │   ├── vector_store.py
    │   └── rag_pipeline.py
    ├── notebooks/
    │   └── 01-rag-experiment.ipynb
    ├── data/
    │   └── sample_documents/
```

```
    ├── tests/
    └── outputs/
```

Do not commit real API keys, private client data, large datasets, or large model files. Use `.env.example`, `.gitignore`, and document where data/models should be stored; GitHub and ML best-practice guidance both recommend keeping secrets and large data out of Git.[1–4][1_1] [^1_3]

4. Root README template

```
# GenAI, ML & Deep Learning Journey

A practical learning repository covering Machine Learning, Deep Learning,
Generative AI, and AI-powered mobile development.

I am a mobile developer with 12 years of experience, currently building
skills in applied AI, LLMs, RAG, agents, and on-device machine learning.

## What This Repository Contains

- ML and Deep Learning concepts
- Interview questions and answers
- Jupyter Notebook experiments
- GenAI POCs: RAG, agents, embeddings, prompt engineering
- Mobile AI examples using Android, iOS, TensorFlow Lite, MediaPipe, and Gemini
- Project case studies and architecture diagrams

## Repository Structure

| Folder | Purpose |
|---|---|
| `01-python-for-ai` | Python, NumPy, Pandas, notebooks |
| `03-machine-learning` | Classical ML concepts and experiments |
| `04-deep-learning` | Neural networks, CNNs, Transformers, PyTorch |
| `05-generative-ai` | LLMs, RAG, agents, embeddings, evaluation |
| `06-mobile-ai` | AI for Android and iOS |
| `07-pocs` | Practical proof-of-concept projects |
| `08-interview-prep` | Interview questions and system design |

## Setup

```bash
python -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```
```

Current Focus

- Python for AI
- ML fundamentals
- Deep Learning with PyTorch

- GenAI: RAG, agents, and LLM applications
- Mobile AI and on-device inference

Notes

This repository is for learning and portfolio demonstration.
No private client data, API keys, or production credentials are committed.

```
A README should explain what the project does, why it is useful, how to get started,

## 5. Suggested first commits

Create the repository in this order:

1. Create `genai-ml-dl-journey` on GitHub.
2. Add `README.md`, `.gitignore`, `LICENSE`, and `requirements.txt`.
3. Add the folder structure above.
4. Add `00-roadmap/learning-roadmap.md`.
5. Add your first Python/ML notebook.
6. Add one ML interview-questions file.
7. Build your first small GenAI POC, such as a document Q&A RAG demo.
8. Add one mobile-AI POC that uses your Android experience.

## 6. Best portfolio strategy

Keep this as your main learning repository, but later create **2-3 polished standalo

- `rag-document-assistant`
- `android-ai-review-analyzer`
- `mobile-voice-agent`

Your learning repo proves breadth and consistency; standalone polished repos prove y

<span style="display:none">[^1_10][^1_11][^1_12][^1_13][^1_14][^1_15][^1_16][^1_17][

<div align="center">⁂</div>

[^1_1]: https://docs.github.com/en/repositories/creating-and-managing-repositories/be
[^1_2]: https://docs.github.com/en/enterprise-server@3.14/repositories/managing-your-
[^1_3]: https://learn.microsoft.com/en-us/training/modules/source-control-for-machine
[^1_4]: https://aigrants.in/topics/best-practices-for-machine-learning-github-reposi
[^1_5]: https://apxml.com/courses/intermediate-python-programming-ml/chapter-6-effic
[^1_6]: https://best-practice-and-impact.github.io/qa-of-code-guidance/project_docume
[^1_7]: https://docs.github.com/en/enterprise-cloud@latest/repositories/creating-and-
[^1_8]: https://docs.github.com/en/enterprise-server@3.21/repositories/creating-and-m
[^1_9]: https://docs.github.com/en/issues/planning-and-tracking-with-projects/learnin
[^1_10]: https://docs.github.com/en/repositories/managing-your-repositorys-settings-a
[^1_11]: https://docs.python-guide.org/writing/structure/
[^1_12]: https://github.com/Azure/machine-learning-best-practices/blob/main/README.md
[^1_13]: https://github.com/recodehive/machine-learning-repos/blob/main/README.md
[^1_14]: https://github.com/daverage/readme.md-best-practice
[^1_15]: https://www.kaggle.com/general/589305
[^1_16]: https://www.perplexity.ai/search/55cf1479-b6bb-4858-9fe0-df53a3e5aa09
```

```
[^1_17]: https://www.perplexity.ai/search/da08dd6e-0feb-482c-87c7-7c0743f73a27
[^1_18]: https://www.perplexity.ai/search/17380354-7b73-47ed-93b8-ffc7ba3d3982
[^1_19]: https://www.perplexity.ai/search/86e2cec2-65ac-42c7-a212-02046c6b2a85
[^1_20]: https://www.perplexity.ai/search/adcae15b-f3cd-4a0c-a5e2-91ca22677e2b

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```